

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

High-Frequency Market Data Processor | C++, Boost Libraries, Custom Memory Pool, Lock-Free Queues[Month] 2026

- Engineered low-latency market data processing engine with custom memory allocators and lock-free data structures, achieving [X] ns average processing time per message (benchmark against system clock)
- Implemented multi-threaded order book reconstruction with NUMA-aware thread pinning and CPU cache optimization, supporting [X]K messages/second throughput (stress test with synthetic data generator)
- Built real-time technical indicator calculations (SMA, EMA, VWAP) using SIMD instructions and template metaprogramming, reducing computation overhead by [X]% (compare with naive implementation)

Crime Analytics & Forecasting Platform | Python, Streamlit, Prophet, Scikit-learn, GeoPandas December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

Real-Time Analytics Pipeline | Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose[Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

TECHNICAL SKILLS

Systems Languages: C++, C, Python, R, SQL (PostgreSQL), MATLAB

Financial Systems: High-Frequency Trading, Market Data Processing, Order Book Management, Technical Indicators

Quantitative Analysis: Time-Series Forecasting, Statistical Modeling, Prophet, Scikit-learn, Pandas, NumPy

Performance Optimization: Custom Memory Allocators, Lock-Free Data Structures, SIMD Instructions, NUMA Optimization

Real-Time Processing: Apache Kafka, Apache Flink, InfluxDB, Stream Processing, Event-Driven Architecture

Data Visualization: Plotly, D3.js, Folium, Geospatial Analysis, Real-Time Dashboards